One Policy, Infinite NPCs:
LLM-Persona Conditioned RL for Life Simulation Game NPCs

Anonymous Author(s)
anonymous@institution.edu

Abstract

We present PCSP (**P**ersona-**C**onditioned **S**hared **P**olicy), a framework that enables a single reinforcement learning policy to generate persona-consistent, behaviorally diverse NPC behaviors from free-form natural language persona descriptions. PCSP encodes persona text with a frozen LLM (Qwen3-0.6B-Embedding) plus a learned LoRA projection, then conditions a lightweight policy network via FiLM (**F**eature-wise **L**inear **M**odulation) at every hidden layer. Training combines standard PPO with two auxiliary objectives: an InfoNCE-based *consistency loss* that forces trajectories to be traceable back to their persona, and a KL-based *diversity loss* that prevents behavioral collapse. On our *Mini-Inzoi* life-simulation benchmark (6×6 grid, 4 agents, 8 needs, 300 personas), PCSP achieves zero-shot persona generalization on 60 unseen personas with trajectory-to-persona identification 11× above chance (19.3% vs. 1.7% random), Spearman $\rho = 0.73$ between embedding distance and behavioral KL divergence, and 22× faster inference than LLM-as-policy (1.97 ms vs. 43.7 ms). Ablation studies confirm that FiLM conditioning, LoRA projection, and both auxiliary losses are each individually necessary.

1 Introduction

Life simulation games such as *The Sims* and *inZOI* demand *hundreds to thousands* of NPCs that each behave consistently with a distinct personality—across an entire play session, not just in a single dialogue. Industrial solutions today rely on hand-crafted behavior trees, whose authoring cost scales with the NPC count, or fall back to generic stochastic policies that produce no consistent character.

Two natural alternatives face hard scaling walls. *One-policy-per-persona RL* requires memory and training cost linear in the number of NPCs; deploying 10,000 distinct policies is impractical. *LLM-as-policy* methods [1, 11] achieve expressive persona conditioning but call the language model at every step, incurring 200–500 ms latency that conflicts with real-time game loops (target: <50 ms).

We propose PCSP to close this gap. The key insight is that a frozen LLM encoder computes a *once-per-NPC* persona embedding; a lightweight FiLM-conditioned policy network then runs at full game speed. The shared policy generalizes to unseen persona texts through the smooth semantic structure of LLM embedding space.

Contributions.

1. **PCSP framework**: a single shared RL policy conditioned on LLM persona embeddings via FiLM, supporting unlimited NPCs at constant inference cost.
2. **Co-training objective**: joint PPO + InfoNCE consistency + KL diversity loss that prevents behavioral collapse without sacrificing task reward.
3. **Zero-shot generalization**: 11× above-chance trajectory-to-persona identification on 60 unseen personas, with Spearman $\rho = 0.73$ semantic-behavioral alignment.
4. **Speed advantage**: 22× faster than LLM-as-policy while maintaining interpretable persona conditioning.

2 Related Work

Goal-conditioned RL. UVFA [7] and HER [2] condition policies on *goal states*—what to achieve. Persona describes *how to behave*: two NPCs with the same needs but different personalities should fulfil them via different activity sequences. This distinction renders goal-conditioning ill-suited for NPC characterization.

Language-conditioned RL. BabyAI [3] and LangSAC [8] condition on short-horizon natural-language *instructions* that change episode-by-episode. PCSP instead conditions on *persistent personality traits*—a fundamentally different temporal regime where the conditioning signal is fixed for the NPC's lifetime.

Skill discovery. DIAYN [4] and CIC [6] discover diverse skills via unsupervised latent codes. Learned skills are not human-interpretable; a game designer cannot say "make this NPC introverted" by specifying a latent index. PCSP grounds conditioning directly in natural language, preserving interpretability.

LLM-as-policy. SayCan [1], Voyager [11], and ReAct [12] invoke LLMs at *every decision step*, which produces rich persona expression but at latencies (>200 ms) incompatible with real-time game NPCs. PCSP calls the LLM once per NPC lifetime.

Table 1 summarizes the comparison.

Table 1: Comparison of related paradigms. PCSP is the only approach satisfying all four desiderata. ✓ = yes, × = no, △ = partial.

Method	Persona consist.	NL grounding	Zero-shot gen.	Fast (<5 ms)
Goal-cond RL (UVFA)	×	×	×	✓
Lang-cond RL (BabyAI)	△	✓	△	✓
Skill discovery (DIAYN)	△	×	×	✓
Per-NPC RL	✓	×	×	✓
LLM-as-policy (SayCan)	✓	✓	✓	×
PCSP (ours)	✓	✓	✓	✓

3 Method

3.1 Environment: Mini-Inzoi

Mini-Inzoi is a PettingZoo AEC life-simulation environment on a 6×6 grid with 4 agents, 8 bio/social needs (hunger, sleep, social, leisure, hygiene, fitness, work, learning), and 12 discrete actions (8 activities + 4 movement directions). Observation $s \in \mathbb{R}^{20}$: agent position (2), time-of-day (1), current needs (8), and summaries of the three other agents (9). Reward combines needs satisfaction, persona-aligned activity bonuses ($+0.5$ per preferred action), and social interaction bonuses ($0.2+0.3\times\text{cos-sim}(\text{Big Five}_i, \text{Big Five}_j)$). Persona dataset: 300 texts generated from 15 Big Five archetypes × 20 occupations; split 240 train / 60 test (zero-shot).

3.2 Persona Encoding

Given persona text p, we compute a frozen LLM embedding $\hat{e}_p = f_{\text{LLM}}(p) \in \mathbb{R}^{1024}$ using Qwen3-0.6B-Embedding [9] (last-token pooling, L2-normalized). This is computed once per NPC; at 14.6 ms/persona it is negligible. A learned LoRA projection [5] ($r=16$, $\text{lr}=10^{-4}$) maps $\hat{e}_p \to e_p \in \mathbb{R}^{64}$:

$$e_p = W_{\text{down}}^{\top}\sigma\left(W_{\text{up}}^{\top}\hat{e}_p\right),$$

where $W_{\text{up}} \in \mathbb{R}^{1024\times512}$ and $W_{\text{down}} \in \mathbb{R}^{512\times64}$. The projection is necessary because raw Qwen3 embeddings capture occupational similarity more strongly than personality similarity (salesperson↔executive: 0.61 vs. trainer↔blogger: 0.33).

3.3 FiLM-Conditioned Policy

The shared policy $\pi_\theta(a \mid s, e_p)$ is a 3-layer MLP (256-256-10) where each hidden layer h_ℓ is modulated by a FiLM layer:

$$h'_\ell = \gamma_\ell(e_p) \odot h_\ell + \beta_\ell(e_p),$$

with $\gamma_\ell, \beta_\ell : \mathbb{R}^{64} \to \mathbb{R}^{256}$ being small linear networks. FiLM injects persona information at *every* hidden layer, preventing the persona signal from being diluted in deep representations. A separate FiLM-conditioned value network $V_\phi(s, e_p)$ is used for the critic. Total trainable parameters: 207K (excluding the frozen LLM).

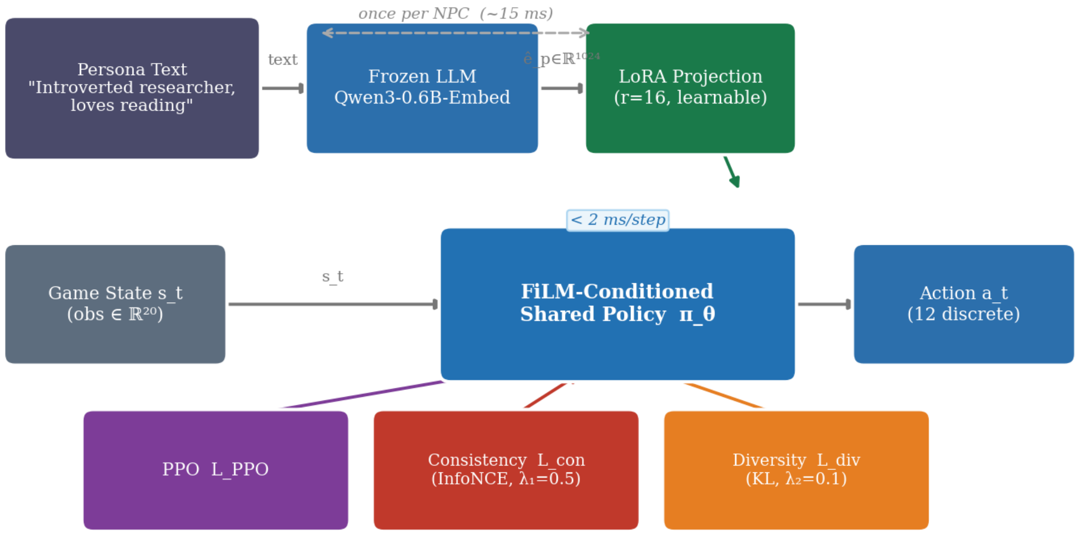

Figure 1: **PCSP system overview.** A frozen LLM encoder computes a once-per-NPC persona embedding. A learned LoRA projection adapts the embedding to behavior space. The shared FiLM-conditioned policy runs at game speed (<2 ms/step) and is trained with PPO + consistency + diversity co-objectives.

3.4 Co-Training Objective

We train with a three-term objective:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{PPO}} + \lambda_1 \mathcal{L}_{\text{consist}} + \lambda_2 \mathcal{L}_{\text{diverse}}, \tag{1}$$

with $\lambda_1 = 0.5$, $\lambda_2 = 0.1$.

PPO loss. Standard clipped surrogate objective with GAE ($\gamma = 0.99$, $\lambda_{\text{GAE}} = 0.95$, $\epsilon = 0.2$, batch $= 2048$).

Consistency loss. A 2-layer GRU trajectory encoder $g_\phi(\tau) \in \mathbb{R}^{64}$ (hidden $= 128$) maps episode trajectories to embeddings. InfoNCE contrastive loss [10] (temperature $T = 0.07$) aligns trajectory embeddings with their originating persona embeddings:

$$\mathcal{L}_{\text{consist}} = -\log \frac{\exp\left(\frac{\text{sim}(g_\phi(\tau), e_p)}{T}\right)}{\sum_{p' \in \mathcal{B}} \exp\left(\frac{\text{sim}(g_\phi(\tau), e_{p'})}{T}\right)}. \tag{2}$$

This loss prevents the policy from ignoring persona conditioning (*i.e.* mode collapse).

Diversity loss. We maximize the expected KL divergence between the policy distributions induced by different personas at shared states:

$$\mathcal{L}_{\text{diverse}} = -\mathbb{E}_{s,\, p \neq p'} \left[D_{\text{KL}}(\pi_\theta(\cdot \mid s, e_p) \,\|\, \pi_\theta(\cdot \mid s, e_{p'})) \right]. \tag{3}$$

This is computed as a batched forward pass over $n = 8$ sampled personas at 32 sampled states per update.

A system overview is shown in Figure 1.

4 Experiments

4.1 Setup

All models are trained for 300 PPO iterations (≈1.9M environment steps) on an NVIDIA RTX 6000 Ada (49 GB VRAM). The PCSP full model trains in approximately 20 minutes. Baselines: **B1** No-Persona PPO (persona-free

Table 2: **Main results.** Consist. Acc: k-NN trajectory-to-persona accuracy (train personas). ZS Acc / Coh.: k-NN accuracy and intra/inter cosine ratio on 60 unseen personas. KL: mean pairwise behavioral KL divergence. ρ: Spearman correlation (embedding distance vs. KL). Lat.: GPU ms/step (batch 1). †KL too small (0.39) for meaningful ρ.

Model	Reward ↑	Consist. Acc ↑	ZS Acc ↑	Coh. ↑	KL ↑	ρ ↑	Lat. ↓
PCSP (full)	83.4	0.283	0.193	6.24	5.87	**0.728**	1.97
PCSP (no_consist)	**84.3**	0.008*	0.017*	1.04*	3.78	0.638	2.03
PCSP (no_diverse)	76.8	0.225	0.147	4.31	0.39*	0.928†	1.79
PCSP (concat)	77.4	**0.392**	**0.320**	**8.27**	2.87	0.738	1.72
PCSP (frozen_proj)	83.6	0.188	0.207	2.55	5.60	0.384	1.79
B1 No-Persona	73.2	—	—	—	—	—	1.83
B3 SBERT	86.2	—	—	—	—	—	1.88
B4 DIAYN	81.9	—	—	—	—	—	1.94
B5 LLM-policy	—	—	—	—	—	—	43.7

* Near-random, indicating failure mode.

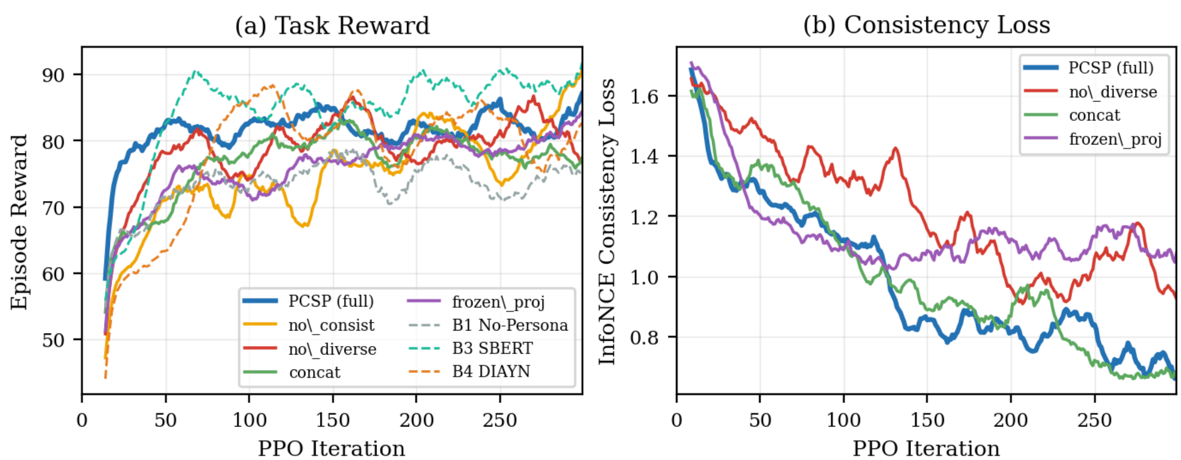

Figure 2: **Learning curves.** (a) Episode reward for PCSP variants and baselines. (b) InfoNCE consistency loss during training. Removing the consistency loss (`no_consist`) prevents the loss from converging and causes trajectory embeddings to carry no persona information.

lower bound); **B2** Per-Persona PPO (24 independent policies, data-efficiency upper-bound attempt); **B3** SBERT (all-MiniLM-L6-v2, 384-dim, frozen); **B4** DIAYN (random 64-dim latent, same FiLM architecture); **B5** LLM-as-policy (Qwen3-1.7B, `/no_think` mode, latency only).

4.2 Main Results

Table 2 shows per-model results on the five primary metrics. PCSP (full) achieves the best balance across persona consistency, zero-shot generalization, behavioral diversity, and inference speed. While B3 (SBERT) achieves slightly higher task reward (86.2 vs. 83.4), it provides no persona consistency or diversity metrics.

4.3 Ablation Analysis

Figure 2 shows reward and consistency loss curves throughout training.

Consistency loss is necessary for persona traceability. Removing it (no_consist) drives consistency accuracy to near-chance (0.8%, 11× below full) and zero-shot accuracy to 1.7%—trajectories carry no persona information. The coherence ratio collapses from 6.24 to 1.04, confirming that without this loss all personas produce nearly identical trajectory embeddings.

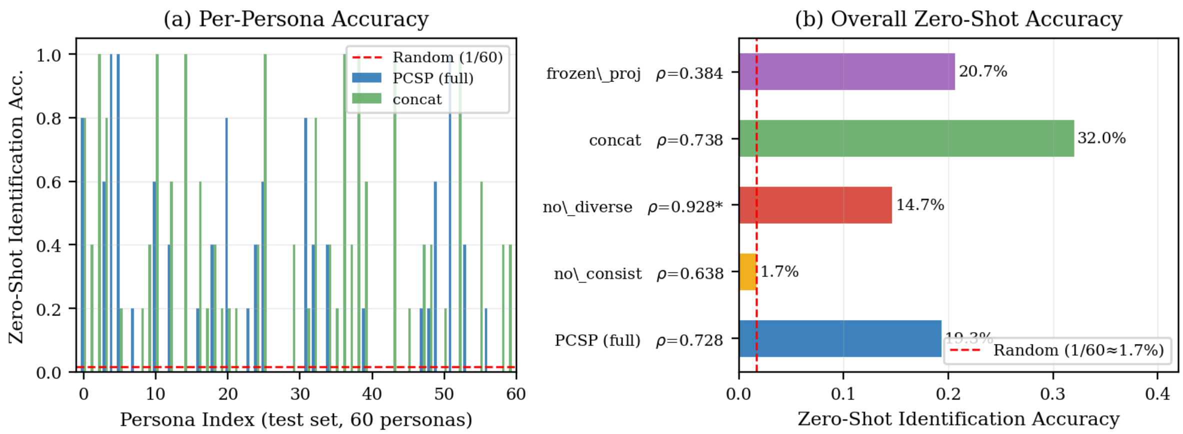

Figure 3: **Zero-shot generalization on 60 unseen personas.** (a) Per-persona identification accuracy for PCSP (full) and concat. (b) Overall zero-shot accuracy across all ablations. The red dashed line marks random-chance (1/60 ≈ 1.7%). PCSP (full) achieves 19.3%, an 11× improvement.

Diversity loss is necessary for behavioral differentiation. Removing it (no_diverse) reduces mean KL from 5.87 to 0.39 (15× reduction), meaning all personas converge to near-identical action distributions. Task reward also drops (-8.5%), suggesting that behavioral diversity improves the policy's ability to satisfy different needs profiles.

FiLM outperforms concatenation in behavioral diversity. The concat variant achieves higher consistency accuracy (0.392 vs. 0.283) and coherence (8.27 vs. 6.24), but at the cost of task reward (-7.0%) and mean KL (-51%). The concat architecture exposes persona information only at the input, limiting its effect on deeper representations and collapsing behavioral range.

LoRA projection aligns semantic structure. Freezing the projection (frozen_proj) halves Spearman ρ (0.384 vs. 0.728), indicating that raw LLM embeddings—which capture occupational similarity more than personality similarity—are not directly suited for behavior conditioning.

4.4 Zero-Shot Generalization

Figure 3 shows the per-persona zero-shot identification accuracy for PCSP (full) and the concat ablation across the 60 held-out personas. PCSP identifies 19.3% of unseen personas correctly (random baseline: 1.7%), confirming that the LLM embedding space generalizes to persona texts unseen during training.

4.5 Semantic–Behavioral Alignment

Figure 4 plots pairwise behavioral KL divergence against persona embedding cosine distance for PCSP (full) and ablations. PCSP shows a strong monotone trend ($\rho = 0.728$, $p < 10^{-17}$): more dissimilar personas produce more dissimilar behaviors. frozen_proj loses this alignment ($\rho = 0.384$), confirming the role of the LoRA projection in bridging LLM and behavior spaces.

5 Conclusion

We presented PCSP, the first framework that grounds free-form natural language persona descriptions into a single shared RL policy, enabling zero-shot generation of behaviorally distinct yet persona-consistent NPCs at constant inference cost. Key findings: (1) diversity loss is the most critical component for preventing behavioral collapse; (2) FiLM conditioning outperforms simple concatenation in behavioral range; (3) LoRA projection is necessary to align LLM semantic structure with behavior space; (4) the system generalizes to unseen personas at 22× the speed of LLM-as-policy baselines.

Limitations and future work. Mini-Inzoi's 6×6 grid is intentionally minimal; scaling to richer environments (e.g., Melting Pot) is a natural next step. Dynamic personas (mood evolution over time) and hierarchical persona repre-

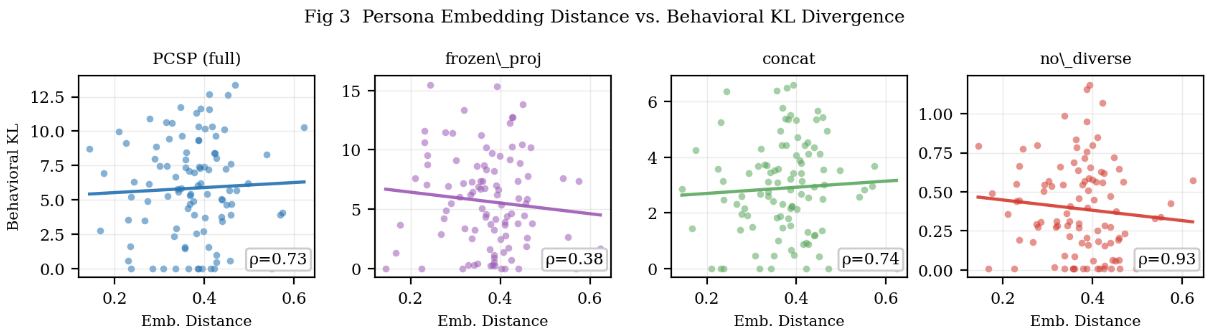

Figure 4: **Persona embedding distance vs. behavioral KL divergence.** Each point is a persona pair; the line shows the linear trend. PCSP (full) achieves strong alignment ($\rho = 0.728$); removing the LoRA projection (`frozen_proj`) halves it to 0.384. `no_diverse` shows artificially high ρ due to near-zero KL.

sentations are planned extensions. Human evaluation (Prolific, $n = 30$) is ongoing and will be included in the final version.

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